

RS001 Week 15 — English Worksheet

Topic: Logic, Knowledge, and Multiple Endings · **Course:** Text Adventure (Python) · **Time:** about 45 minutes

This week the ending depends on what the player **knows**, not on luck. In your live lesson you built code that stores riddles and answers in parallel lists, keeps a **score**, and uses the score to pick one of several endings. This worksheet is about reading and explaining that code in your own words.

 Words to know: **score**, **riddle**, **answer**, **ending**, **logic**

```
questions = ["I have four legs but cannot walk. What am I?"]
answers    = ["table"]
guess = game.ask(questions[0]).strip()
if guess == answers[0]:
    score = score + 1
```

1 · Predict

Read each piece of code. Work it out in your head and write what you think happens. Do not run it.

```
score = 0
score = score + 1
score = score + 1
game.say(score)
```

What number is printed?


```
questions = ["cold and melts fast?", "rains and it opens?"]
answers    = ["ice", "umbrella"]
guess = "ice"
if guess == answers[0]:
    game.say("correct")
else:
    game.say("wrong")
```

Which message appears?


```
score = 2
if score == 3:
    game.say("perfect")
elif score == 2:
    game.say("great")
else:
    game.say("keep trying")
```

Which ending prints, and why does the first branch get skipped?

2 · Spot the Bug 🐛

Each block was meant to do something, but it is broken. Read what it is **supposed** to do, fix it on paper, then explain why the original was wrong.

Bug A — A correct answer should add **one** to the score. Right now the score never goes up, because the result of the addition is thrown away.

```
score = 0
guess = "table"
answer = "table"
if guess == answer:
    score + 1
game.say(score)
```

Hint: store the new total back into `score` with `score = score + 1`.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The riddle and its answer should share the **same** index. Here question 0 is checked against answer 1, so a correct guess is marked wrong.

```
questions = ["four legs, cannot walk?", "cold, melts fast?"]
answers   = ["table", "ice"]
guess = game.ask(questions[0]).strip()
if guess == answers[1]:
    game.say("correct")
else:
    game.say("wrong")
```

Hint: match the answer index to the question index.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The endings should split by score: 3 = wise, 2 = hero, else = beginner. Right now every score over 0 gives the same ending, because separate `if` s overwrite each other.

```
score = 3
if score >= 1:
    ending = "beginner"
if score >= 2:
    ending = "hero"
if score == 3:
    ending = "wise"
game.say(ending)
```

Hint: use `if` / `elif` / `else` and check the **highest** score first.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Explain the Code

This is a working version of the score-based ending code, like the one from your lesson. Read it carefully, then answer the questions below.

```
questions = ["four legs, cannot walk?", "cold, melts fast?", "opens when it rains?"]
answers    = ["table", "ice", "umbrella"]
score = 0

for i in range(len(questions)):
    guess = game.ask(questions[i]).strip()
    if guess == answers[i]:
        score = score + 1

if score == 3:
    ending = "wise"
elif score == 2:
    ending = "hero"
else:
    ending = "beginner"

game.say("Your ending: " + ending)
```

Why do **questions** and **answers** need to be the same length?

What does **answers[i]** give you on the second loop, when **i** is 1?

What is the largest number score can reach with this code, and why?

A player gets 2 riddles right. Which ending do they see, and which branch is checked first?

Why does this game give a fair ending based on knowledge, not luck?

4 · Explain Your Lesson Code 🎥

Take a short video on your phone (or a parent's phone) explaining the score-based ending code **you** wrote in today's lesson. You may show it running. Teach it like the viewer has never coded. Try to use these words: **score, riddle, answer, ending, logic**.

1. Show your riddle and answer lists, and explain how they line up.
2. Point to the line where the score goes up, and say when it happens.
3. Show the different endings and explain what decides which one appears.
4. Explain why the ending depends on knowledge, not luck.

Write what you will say in your video. Plan it here before you record.

Submit 

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.